SAEON Agile Logbook

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Field	Record
Student number	222641681
Role	Project Leader and hardware integration lead
Project	SAEON Real-Time Rainfall Monitoring System
Reporting period	28 July to 23 October 2026
Working days	Tuesday to Friday

Purpose and project configuration

This individual logbook records progress, planned next steps and impediments against the project's 13-week schedule. Weekly headings identify the planned work area; the entries record individual contributions, including supporting work outside a person's main role.

The ESP32 WROOM-32 reads the TR-525M rain gauge. DHT11 is used for testing; SHT31 is the temperature and humidity sensor selected for the final product. Firebase is the database platform.

The agreed architecture retains SIM7000G cellular communication and HiveMQ Cloud MQTT, with a backend ingestion service writing to Firebase. Final deployment and field validation remain open.

Record status

The daily entries are reconstructed working records and require confirmation by the named student. They are not verified attendance or proof that a task was completed. Impediments describe possible risks unless a dated incident is confirmed. Correct any inaccurate activity before signing the logbook.

This revision confirms the technology choices; it does not verify past daily activities or create missing test results. Proposed current follow-up actions appear after the dated entries.

Daily questions

- What did I complete since the last standup?
- What am I planning to complete between now and the next standup?
- What are my impediments (or risks or problems)?

The tables use shortened labels for these same three questions. Mondays are outside the supplied reporting schedule; no Monday attendance or activity is inferred. Historical stand-up meetings are not implied by daily entries.

Week 1 28 July 2026 to 31 July 2026

Planned work area: Project Initiation & System Architecture
PB01 | US01, US02, US03, US04, US13

Tuesday 28 July 2026

Completed since the last stand-up	I reviewed and documented project scope.
Plan before the next stand-up	I will continue with component verification and record the results.
Impediments, risks or problems	I need the agreed scope and task ownership to remain clear.

Wednesday 29 July 2026

Completed since the last stand-up	I worked on component verification.
Plan before the next stand-up	I will continue with development setup and record the results.
Impediments, risks or problems	I need the agreed scope and task ownership to remain clear.

Thursday 30 July 2026

Completed since the last stand-up	I checked and updated development setup.
Plan before the next stand-up	I will continue with architecture review and record the results.
Impediments, risks or problems	I need the agreed scope and task ownership to remain clear.

Friday 31 July 2026

Completed since the last stand-up	I organised the evidence for architecture review.
Plan before the next stand-up	I will continue with ESP32 base setup and record the results.
Impediments, risks or problems	I need the agreed scope and task ownership to remain clear.

Week 2 4 August 2026 to 7 August 2026

Planned work area: Core Firmware & Sensor Interfacing
PB02 | US01, US02, US11

Tuesday 4 August 2026

Completed since the last stand-up	I reviewed and documented ESP32 base setup.
Plan before the next stand-up	I will continue with temperature sensor integration and record the results.
Impediments, risks or problems	I depend on working sensors and a stable bench setup.

Wednesday 5 August 2026

Completed since the last stand-up	I worked on temperature sensor integration.
Plan before the next stand-up	I will continue with rain-gauge tip counting and record the results.
Impediments, risks or problems	I depend on working sensors and a stable bench setup.

Thursday 6 August 2026

Completed since the last stand-up	I checked and updated rain-gauge tip counting.
Plan before the next stand-up	I will continue with firmware low-power behaviour and record the results.
Impediments, risks or problems	I depend on working sensors and a stable bench setup.

Friday 7 August 2026

Completed since the last stand-up	I organised the evidence for firmware low-power behaviour.
Plan before the next stand-up	I will continue with hardware message requirements and record the results.
Impediments, risks or problems	I depend on working sensors and a stable bench setup.

Sensor context: DHT11 is the testing sensor. SHT31 is selected for the final product; final verification is tracked under PB17.

Week 3 11 August 2026 to 14 August 2026

Planned work area: HiveMQ Cloud Broker Setup
PB03 | US03, US04, US09

Tuesday 11 August 2026

Completed since the last stand-up	I reviewed and documented hardware message requirements.
Plan before the next stand-up	I will continue with station topic requirements and record the results.
Impediments, risks or problems	I need consistent message fields, permissions and timestamps.

Wednesday 12 August 2026

Completed since the last stand-up	I worked on station topic requirements.
Plan before the next stand-up	I will continue with sensor MQTT test support and record the results.
Impediments, risks or problems	I need consistent message fields, permissions and timestamps.

Thursday 13 August 2026

Completed since the last stand-up	I checked and updated sensor MQTT test support.
Plan before the next stand-up	I will continue with hardware-to-storage integration checks and record the results.
Impediments, risks or problems	I need consistent message fields, permissions and timestamps.

Friday 14 August 2026

Completed since the last stand-up	I organised the evidence for hardware-to-storage integration checks.
Plan before the next stand-up	I will continue with SIM7000G wiring and power and record the results.
Impediments, risks or problems	I need consistent message fields, permissions and timestamps.

Week 4 18 August 2026 to 21 August 2026

Planned work area: Cellular Communication (SIM7000G)
PB04 | US03, US04

Tuesday 18 August 2026

Completed since the last stand-up	I reviewed and documented SIM7000G wiring and power.
Plan before the next stand-up	I will continue with network configuration review and record the results.
Impediments, risks or problems	I depend on cellular coverage, an active SIM and stable modem power.

Wednesday 19 August 2026

Completed since the last stand-up	I worked on network configuration review.
Plan before the next stand-up	I will continue with cellular MQTT integration and record the results.
Impediments, risks or problems	I depend on cellular coverage, an active SIM and stable modem power.

Thursday 20 August 2026

Completed since the last stand-up	I checked and updated cellular MQTT integration.
Plan before the next stand-up	I will continue with connection stability checks and record the results.
Impediments, risks or problems	I depend on cellular coverage, an active SIM and stable modem power.

Friday 21 August 2026

Completed since the last stand-up	I organised the evidence for connection stability checks.
Plan before the next stand-up	I will continue with dashboard hardware requirements and record the results.
Impediments, risks or problems	I depend on cellular coverage, an active SIM and stable modem power.

Week 5 25 August 2026 to 28 August 2026

Planned work area: SAEON Dashboard Base Application
PB05 | US05, US06, US09

Tuesday 25 August 2026

Completed since the last stand-up	I reviewed and documented dashboard hardware requirements.
Plan before the next stand-up	I will continue with live-reading integration and record the results.
Impediments, risks or problems	I need the dashboard to handle missing and outdated data clearly.

Wednesday 26 August 2026

Completed since the last stand-up	I worked on live-reading integration.
Plan before the next stand-up	I will continue with historical-data consistency and record the results.
Impediments, risks or problems	I need the dashboard to handle missing and outdated data clearly.

Thursday 27 August 2026

Completed since the last stand-up	I checked and updated historical-data consistency.
Plan before the next stand-up	I will continue with cloud-command preparation boundaries and record the results.
Impediments, risks or problems	I need the dashboard to handle missing and outdated data clearly.

Friday 28 August 2026

Completed since the last stand-up	I organised the evidence for cloud-command preparation boundaries.
Plan before the next stand-up	I will continue with battery and charge-controller setup and record the results.
Impediments, risks or problems	I need the dashboard to handle missing and outdated data clearly.

Week 6 1 September 2026 to 4 September 2026

Planned work area: Power Supply & Solar Integration
PB06 | US11

Tuesday 1 September 2026

Completed since the last stand-up	I reviewed and documented battery and charge-controller setup.
Plan before the next stand-up	I will continue with solar panel integration and record the results.
Impediments, risks or problems	I need verified power records before claiming an operating duration.

Wednesday 2 September 2026

Completed since the last stand-up	I worked on solar panel integration.
Plan before the next stand-up	I will continue with alternative power checks and record the results.
Impediments, risks or problems	I need verified power records before claiming an operating duration.

Thursday 3 September 2026

Completed since the last stand-up	I checked and updated alternative power checks.
Plan before the next stand-up	I will continue with power-flow audit and record the results.
Impediments, risks or problems	I need verified power records before claiming an operating duration.

Friday 4 September 2026

Completed since the last stand-up	I organised the evidence for power-flow audit.
Plan before the next stand-up	I will continue with station power diagnostics and record the results.
Impediments, risks or problems	I need verified power records before claiming an operating duration.

Week 7 8 September 2026 to 11 September 2026

Planned work area: Advanced UI & Health Diagnostics
PB07 | US07, US08, US10

Tuesday 8 September 2026

Completed since the last stand-up	I reviewed and documented station power diagnostics.
Plan before the next stand-up	I will continue with sensor units in exported data and record the results.
Impediments, risks or problems	I need diagnostic and exported values to match the stored readings.

Wednesday 9 September 2026

Completed since the last stand-up	I worked on sensor units in exported data.
Plan before the next stand-up	I will continue with dashboard hardware feedback and record the results.
Impediments, risks or problems	I need diagnostic and exported values to match the stored readings.

Thursday 10 September 2026

Completed since the last stand-up	I checked and updated dashboard hardware feedback.
Plan before the next stand-up	I will continue with end-to-end hardware checks and record the results.
Impediments, risks or problems	I need diagnostic and exported values to match the stored readings.

Friday 11 September 2026

Completed since the last stand-up	I organised the evidence for end-to-end hardware checks.
Plan before the next stand-up	I will continue with permanent circuit layout and record the results.
Impediments, risks or problems	I need diagnostic and exported values to match the stored readings.

Week 8 15 September 2026 to 18 September 2026

Planned work area: Hardware Finalization & PCB Soldering
PB08 | US12

Tuesday 15 September 2026

Completed since the last stand-up	I reviewed and documented permanent circuit layout.
Plan before the next stand-up	I will continue with PCB soldering and record the results.
Impediments, risks or problems	I need the permanent assembly to match the reviewed wiring.

Wednesday 16 September 2026

Completed since the last stand-up	I worked on PCB soldering.
Plan before the next stand-up	I will continue with enclosure preparation and record the results.
Impediments, risks or problems	I need the permanent assembly to match the reviewed wiring.

Thursday 17 September 2026

Completed since the last stand-up	I checked and updated enclosure preparation.
Plan before the next stand-up	I will continue with component mounting and sealing and record the results.
Impediments, risks or problems	I need the permanent assembly to match the reviewed wiring.

Friday 18 September 2026

Completed since the last stand-up	I organised the evidence for component mounting and sealing.
Plan before the next stand-up	I will continue with power-drain audit and record the results.
Impediments, risks or problems	I need the permanent assembly to match the reviewed wiring.

Week 9 22 September 2026 to 25 September 2026

Planned work area: Pre-Deployment Auditing & Calibration
PB09 | US11, US12

Tuesday 22 September 2026

Completed since the last stand-up	I reviewed and documented power-drain audit.
Plan before the next stand-up	I will continue with solar-charging verification and record the results.
Impediments, risks or problems	I still depend on completion of installation and baseline field calibration.

Wednesday 23 September 2026

Completed since the last stand-up	I worked on solar-charging verification.
Plan before the next stand-up	I will continue with installation preparations and record the results.
Impediments, risks or problems	I still depend on completion of installation and baseline field calibration.

Thursday 24 September 2026

Completed since the last stand-up	I prepared notes on installation preparations. The field work remains in progress.
Plan before the next stand-up	I will continue with baseline field-calibration preparations and record the results.
Impediments, risks or problems	I still depend on completion of installation and baseline field calibration.

Friday 25 September 2026

Completed since the last stand-up	I prepared notes on baseline field-calibration preparations. The field work remains in progress.
Plan before the next stand-up	I will continue with field-test setup review and record the results.
Impediments, risks or problems	I still depend on completion of installation and baseline field calibration.

Week 10 29 September 2026 to 2 October 2026

Planned work area: 48-Hour Field Stress Test
PB10 | US01, US02, US03, US11, US12

Tuesday 29 September 2026

Completed since the last stand-up	I prepared notes on field-test setup review. The field work remains in progress.
Plan before the next stand-up	I will continue with hardware monitoring support and record the results.
Impediments, risks or problems	I cannot treat field validation as complete while the test and audit remain in progress.

Wednesday 30 September 2026

Completed since the last stand-up	I prepared notes on hardware monitoring support. The field work remains in progress.
Plan before the next stand-up	I will continue with reference-measurement setup and record the results.
Impediments, risks or problems	I cannot treat field validation as complete while the test and audit remain in progress.

Thursday 1 October 2026

Completed since the last stand-up	I prepared notes on reference-measurement setup. The field work remains in progress.
Plan before the next stand-up	I will continue with outstanding field-validation issues and record the results.
Impediments, risks or problems	I cannot treat field validation as complete while the test and audit remain in progress.

Friday 2 October 2026

Completed since the last stand-up	I prepared notes on outstanding field-validation issues. The field work remains in progress.
Plan before the next stand-up	I will continue with hardware presentation outline and record the results.
Impediments, risks or problems	I cannot treat field validation as complete while the test and audit remain in progress.

Week 11 6 October 2026 to 9 October 2026

Planned work area: Presentation Formatting & Content
PB11 | US13

Tuesday 6 October 2026

Completed since the last stand-up	I reviewed and documented hardware presentation outline.
Plan before the next stand-up	I will continue with wiring and architecture diagrams and record the results.
Impediments, risks or problems	I need the presentation to distinguish prototype results from pending field results.

Wednesday 7 October 2026

Completed since the last stand-up	I worked on wiring and architecture diagrams.
Plan before the next stand-up	I will continue with hardware details in dashboard slides and record the results.
Impediments, risks or problems	I need the presentation to distinguish prototype results from pending field results.

Thursday 8 October 2026

Completed since the last stand-up	I checked and updated hardware details in dashboard slides.
Plan before the next stand-up	I will continue with wiring and power guide and record the results.
Impediments, risks or problems	I need the presentation to distinguish prototype results from pending field results.

Friday 9 October 2026

Completed since the last stand-up	I organised the evidence for wiring and power guide.
Plan before the next stand-up	I will continue with hardware slide review and record the results.
Impediments, risks or problems	I need the presentation to distinguish prototype results from pending field results.

Week 12 13 October 2026 to 16 October 2026

Planned work area: Rehearsal & Submission Prep
PB12 | US13

Tuesday 13 October 2026

Completed since the last stand-up	I reviewed and documented hardware slide review.
Plan before the next stand-up	I will continue with hardware demonstration rehearsal and record the results.
Impediments, risks or problems	I need a reliable demonstration and clear documentation of open field work.

Wednesday 14 October 2026

Completed since the last stand-up	I worked on hardware demonstration rehearsal.
Plan before the next stand-up	I will continue with technical explanation review and record the results.
Impediments, risks or problems	I need a reliable demonstration and clear documentation of open field work.

Thursday 15 October 2026

Completed since the last stand-up	I checked and updated technical explanation review.
Plan before the next stand-up	I will continue with prototype package hardware checks and record the results.
Impediments, risks or problems	I need a reliable demonstration and clear documentation of open field work.

Friday 16 October 2026

Completed since the last stand-up	I organised the evidence for prototype package hardware checks.
Plan before the next stand-up	I will continue with prototype documentation review and record the results.
Impediments, risks or problems	I need a reliable demonstration and clear documentation of open field work.

Week 13 20 October 2026 to 23 October 2026

Planned work area: Project Buffer & Final Review
PB13 | US12, US13

Tuesday 20 October 2026

Completed since the last stand-up	I reviewed and documented prototype documentation review.
Plan before the next stand-up	I will continue with known hardware limitations and record the results.
Impediments, risks or problems	I am still awaiting completion of final deployment and field validation.

Wednesday 21 October 2026

Completed since the last stand-up	I worked on known hardware limitations.
Plan before the next stand-up	I will continue with field-deployment follow-up and record the results.
Impediments, risks or problems	I am still awaiting completion of final deployment and field validation.

Thursday 22 October 2026

Completed since the last stand-up	I checked and updated field-deployment follow-up.
Plan before the next stand-up	I will continue with prototype documentation sign-off and record the results.
Impediments, risks or problems	I am still awaiting completion of final deployment and field validation.

Friday 23 October 2026

Completed since the last stand-up	I organised the evidence for prototype documentation sign-off.
Plan before the next stand-up	I will continue with the remaining deployment and field-validation actions and record the results.
Impediments, risks or problems	I am still awaiting completion of final deployment and field validation.

Taskboard summary and current follow-up

Snapshot retained from the project backlog at 23 October 2026. These shared statuses do not mean that this student personally completed every task.

Product Backlog	In Progress	Completed
PB17: Final SHT31 verification. PB18: Firebase access and ingestion checks. PB19: Assessment evidence package.	9.03 Installation. 9.04 Baseline calibration. 10.01 Test initiation. 10.02 Active monitoring. 10.03 Reference measurements. 10.04 Audit and conclusions.	43 original task cards reported complete. This covers planning, prototype development and bench work, documentation and presentation preparation. Evidence checks remain necessary.
Optional: PB14 pressure, wind and UV; PB15 GPS; PB16 over-the-air updates.	Field acceptance remains open.	Prototype documentation sign-off is separate from field acceptance.

What do we need to do to advance this piece of work

Confirm the SHT31 final configuration and Firebase evidence, finish installation and baseline calibration, then complete and audit the 48-hour field test.

Is anyone working on anything that is not on the board

Ethan's 4–7 August entries include simulation work, HTTP connection investigation and a supervisor meeting about a pre-seed presentation. Cross-reference these as historical support activities; do not replace them with planned sprint tasks. Record any further unlisted work when confirmed.

What do we need to finish as a team

Close the six open field tasks, confirm PB17–PB19 evidence, verify individual daily entries and complete the graduate-attribute and GitHub portfolio evidence.

Are there any bottlenecks or blockers to the flow of work

Open dependencies include installation, calibration, final-sensor evidence and field-test records. Site access, modem power and connectivity are risks to check; do not record them as incidents without dated evidence.

My next action

Coordinate final SHT31 verification, installation and calibration, and confirm the hardware evidence.